

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Data Visualization with Power BI and Tableau

Practical – V

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Aim:-Analyzing Data using Table Calculations in Tableau

a) Apply quick table calculations (running total, percent of total)

Steps

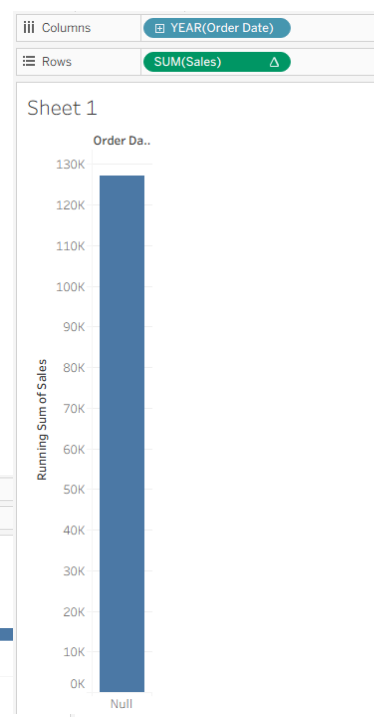
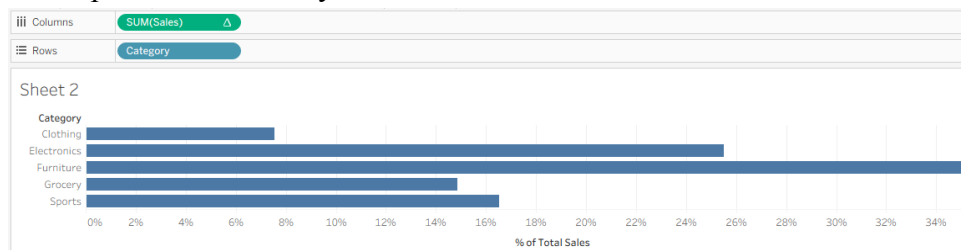
1. Open a new worksheet.
2. Drag Order_Date to the Columns shelf.
3. Drag Sales to the Rows shelf.
4. Right-click SUM(Sales).
5. Select Quick Table Calculation → Running Total.

Observation

The chart displays the cumulative sales over time by continuously adding the current sales value to the previous total.

Why We Are Doing It

Running Total helps analyze cumulative sales growth over a period and identify overall business trends.



b) Create moving averages and cumulative metrics

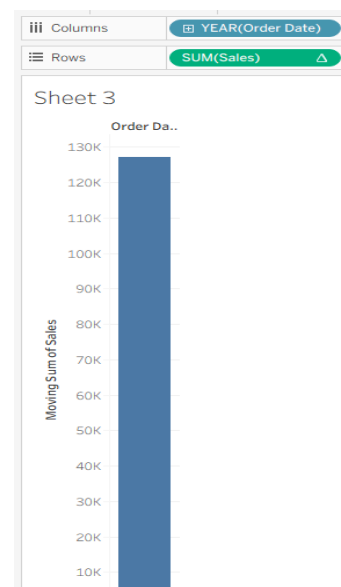
Steps

1. Open a new worksheet.
2. Drag Category to the Rows shelf.
3. Drag Sales to the Columns shelf.
4. Right-click SUM(Sales).
5. Select Quick Table Calculation → Percent of Total.
6. Format the values as Percentage.

Observation

Each category displays its percentage contribution to the total sales.

Why We Are Doing It



Percent of Total helps compare how much each category contributes to the overall sales.

c) Perform year-over-year growth analysis

Steps

1. Open a new worksheet.
2. Drag YEAR(Order_Date) to the Columns shelf.
3. Drag Sales to the Rows shelf.
4. Right-click SUM(Sales).
5. Select Quick Table Calculation → Percent Difference.
6. Right-click SUM(Sales) again.
7. Select Compute Using → Table (Across).

Observation

The worksheet shows the percentage increase or decrease in sales compared with the previous year.

Why We Are Doing It

Year-over-Year Growth helps compare annual business performance.

d) Use ranking functions (Top N, Bottom N)

Steps

1. Right-click the Data Pane.
2. Select Create → Calculated Field.
3. Name the calculated field Sales Rank.
4. Enter the formula:

`RANK(SUM([Sales]), 'desc')`

5. Click OK.
6. Drag Category to the Rows shelf.
7. Drag Sales to the Columns shelf.
8. Drag Sales Rank beside Category.

Observation

Each category receives a rank according to its total sales.

Rank 1 represents the highest sales.

Why We Are Doing It

Ranking helps identify the Top-performing and Bottom-performing categories.

e) Implement window functions (WINDOW_SUM, WINDOW_AVG)

Steps

1. Create a calculated field named Window Sum Sales.
2. Enter:

`WINDOW_SUM(SUM([Sales]))`

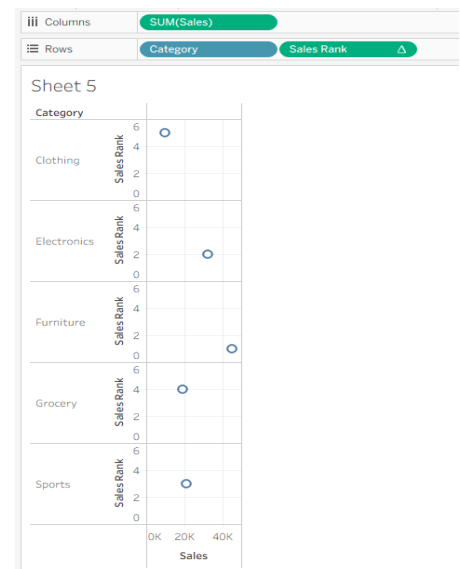
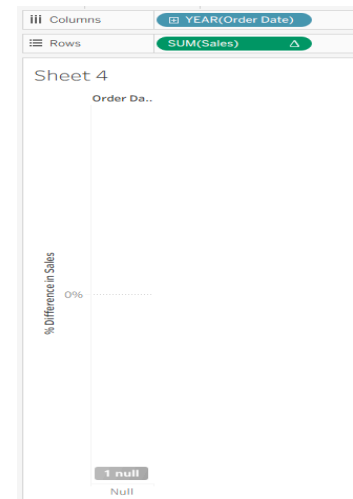
3. Click OK.
4. Drag State to the Rows shelf.
5. Drag Window Sum Sales to the Columns shelf.

Observation

The calculation displays the total sales for the current window of data.

Why We Are Doing It

WINDOW_SUM calculates totals within the current table instead of the entire dataset.



WINDOW_AVG

Steps

1. Create a calculated field named Window Average Sales.
2. Enter:

WINDOW_AVG(AVG([Sales]))

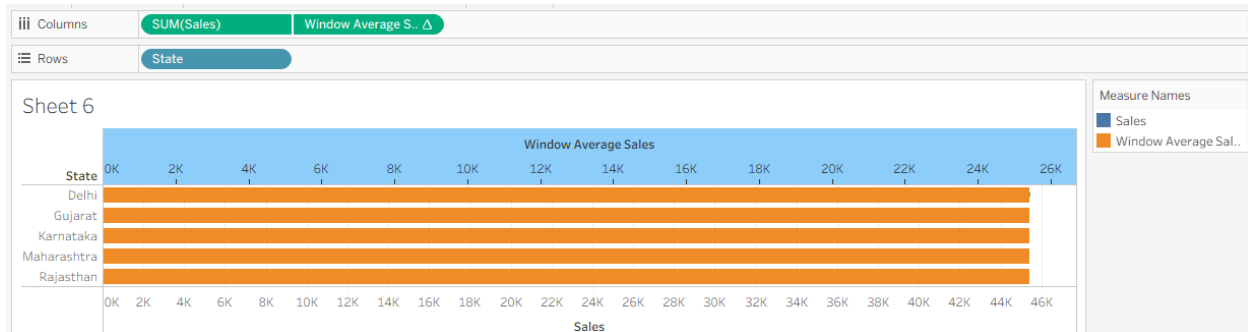
3. Click OK.
4. Drag State to the Rows shelf.
5. Drag Sales to the Columns shelf.
6. Drag Window Average Sales to the Columns shelf beside Sales.
7. Right-click Window Average Sales and select Dual Axis.
8. Right-click one of the axes and select Synchronize Axis.

Observation

The bar chart displays sales for each state, while the line shows the average sales across all states.

Why We Are Doing It

WINDOW_AVG compares each state's sales with the average sales in the current view.



f) Customize addressing and partitioning

Steps

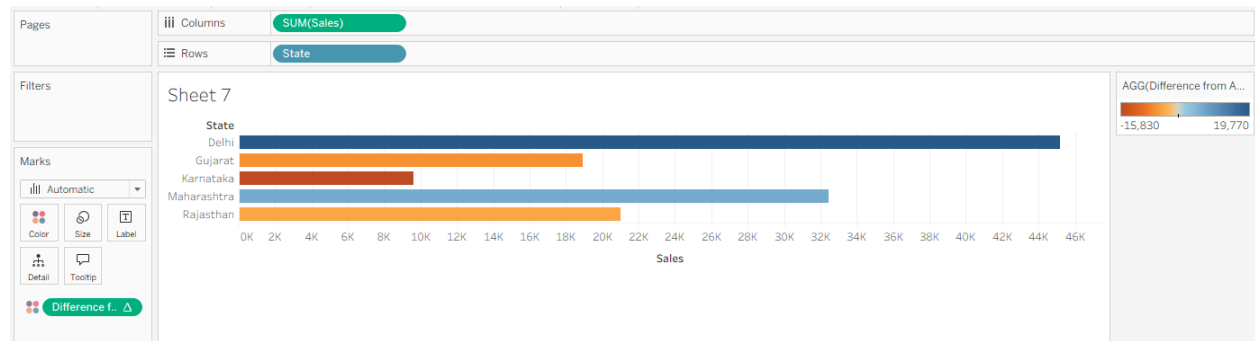
1. Open a new worksheet.
2. Drag State to the Rows shelf.
3. Drag Category next to State.
4. Drag Sales to the Columns shelf.
5. Right-click SUM(Sales).
6. Select Quick Table Calculation → Rank.
7. Right-click the measure again and choose Edit Table Calculation.
8. Change Compute Using between:
 - Table (Down)
 - Pane (Down)

Observation

The ranking changes depending on the selected computation direction.

Why We Are Doing It

Addressing and Partitioning determine how Tableau performs table calculations across rows and panes.



g) Create dynamic titles and labels using table calculations

Steps

1. Create a parameter named Top N Threshold.
2. Set:
 - Data Type: Integer
 - Current Value: 5
3. Show the parameter.
4. Create a calculated field named Dynamic Title.
5. Enter:

"Top " + STR([Top N Threshold]) + " Categories by Sales"

6. Edit the worksheet title.
7. Click Insert → Parameters → Top N Threshold.
8. Set the title as:

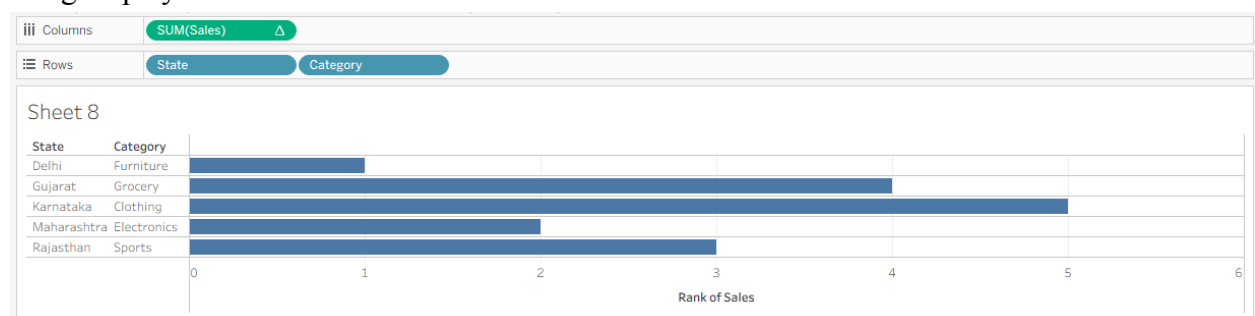
Top <Parameters.Top N Threshold> Categories by Sales

Observation

The worksheet title changes automatically whenever the parameter value changes.

Why We Are Doing It

Dynamic titles make the worksheet more interactive and clearly indicate the current analysis being displayed.



Conclusion

This practical demonstrated the use of Table Calculations in Tableau to analyze sales data effectively. Running totals, percentages, moving averages, year-over-year growth, ranking, and window functions were used to gain meaningful insights, while addressing, partitioning, and dynamic titles enhanced the interactivity and presentation of the visualizations. Overall, the practical improved understanding of advanced data analysis techniques in Tableau.